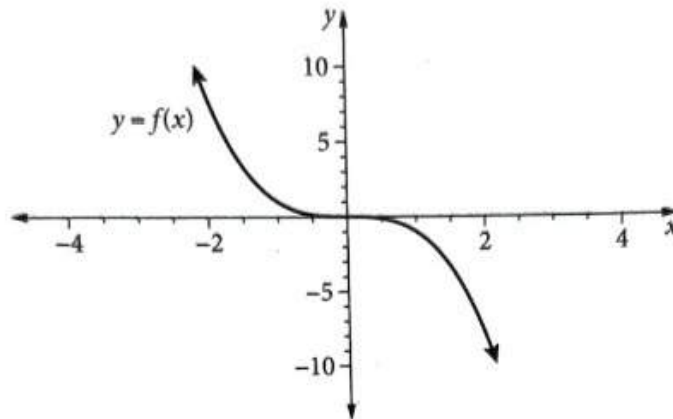
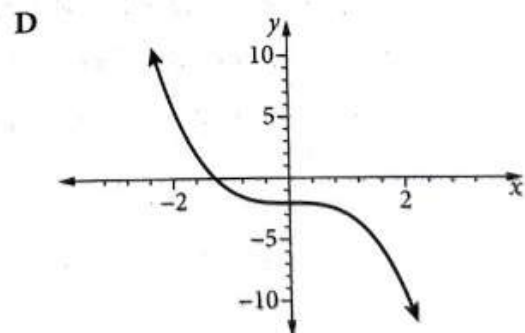
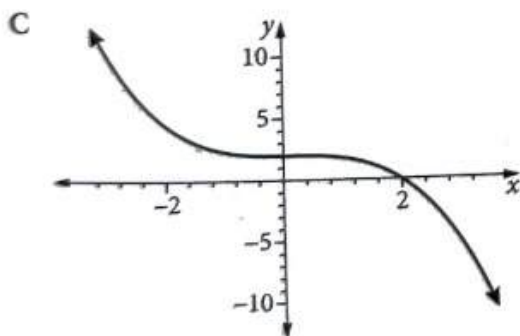
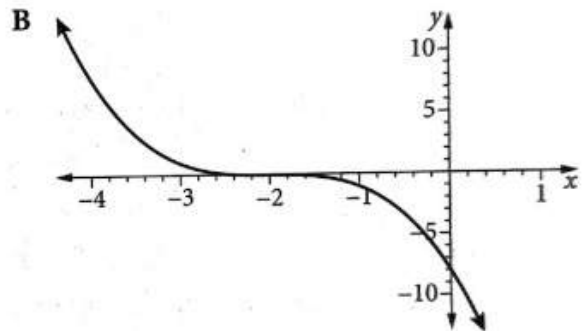
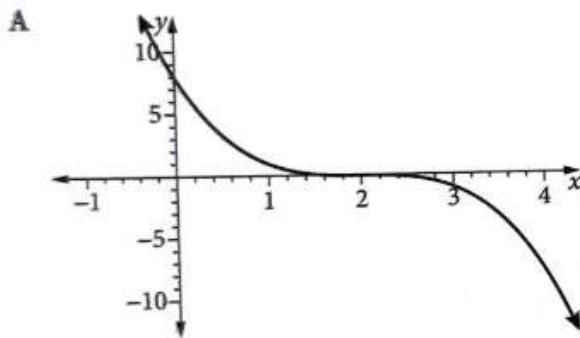


Question 1

The graph of $y = f(x)$ is shown.



Which of the following is the graph of $y = f(x) - 2$?



Question 2

If the point $(6, 2)$ is translated 4 units left and then reflected in the x -axis, what are its new coordinates?

A $(2, -2)$

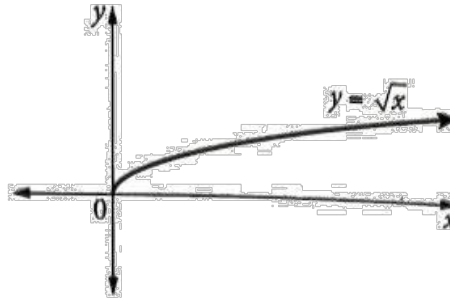
B $(0, 4)$

C $(-10, 2)$

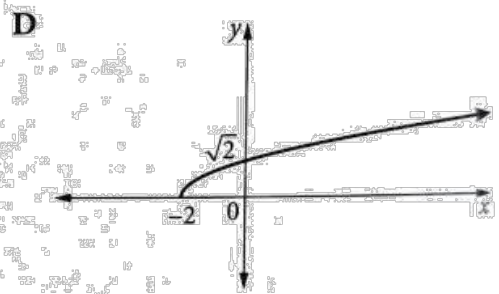
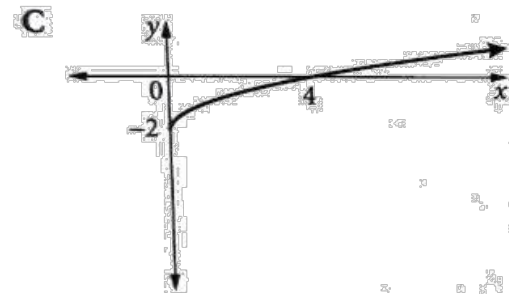
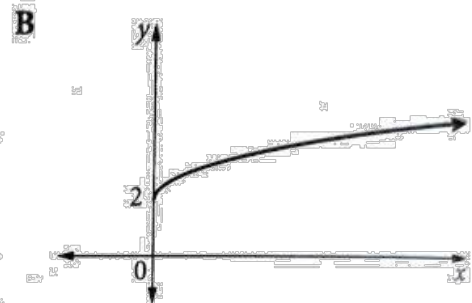
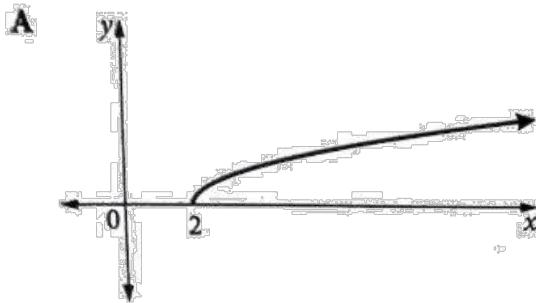
D $(-6, -2)$

Question 3

The graph of $y = \sqrt{x}$ is shown below.



Which of the following is the graph of $y = \sqrt{x+2}$?



Question 4

Which sequence of transformations changes the graph from $y = x^3$ to $y = (x-5)^3 + 2$?

- A** Horizontal translation 5 units to the right and a vertical translation 2 units up
- B** Horizontal translation 5 units to the left and a vertical translation 2 units up
- C** Horizontal translation 2 units to the left and a vertical translation 5 units down
- D** Horizontal translation 2 units to the right and a vertical translation 5 units down

Question 5

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Which inequality gives the domain of $y = \sqrt{2x-3}$?

A $x < \frac{3}{2}$

B $x > \frac{3}{2}$

C $x \leq \frac{3}{2}$

D $x \geq \frac{3}{2}$

Question 6

The graph of the function $y = f(x)$ passes through the point $(-3, 8)$. If $g(x) = f\left(\frac{x}{3}\right) + 4$, then through which of the following points does the graph of $y = g(x)$ pass?

A $(-1, 12)$

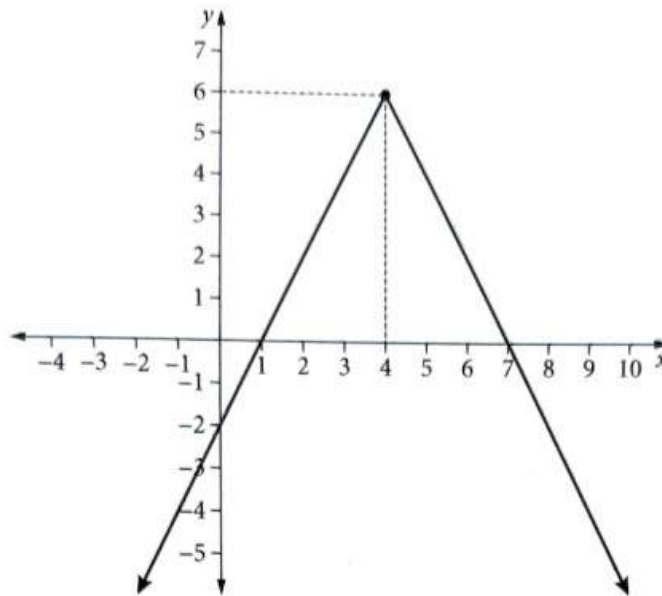
B $(-1, 28)$

C $(-9, 12)$

D $\left(1, \frac{8}{3}\right)$

Question 7

The function $f(x) = |x|$ is transformed and the graph of the new function is shown below.



The equation of the new function is $y = k|x + b| + c$. What are the values of k , b and c ?

A $k = 2$, $b = -6$ and $c = 4$

B $k = -2$, $b = -4$ and $c = 6$

C $k = 2$, $b = 4$ and $c = -6$

D $k = -2$, $b = 6$ and $c = 4$

Question 8

The graph of $y = \frac{5}{x+3}$ is translated 4 units right and dilated vertically by a factor of $\frac{1}{2}$.

What is the equation of the transformed graph?

A $y = \frac{10}{x-1}$

B $y = \frac{5}{2(x-7)}$

C $y = \frac{5}{2(x-1)}$

D $y = \frac{10}{x-7}$

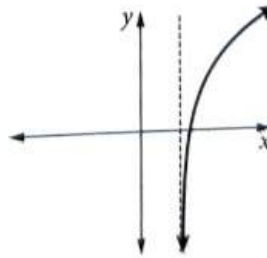
Question 9

Which one of the following graphs could represent the transformation of $f(x) = \log_3 x$ to the graph of $f(x) = \log_3(x - p) + q$, where p and q are both positive?

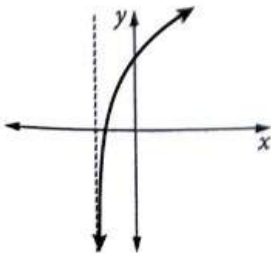
A



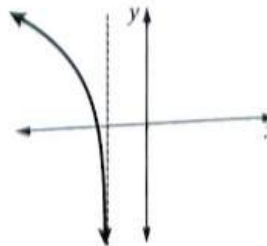
B



C

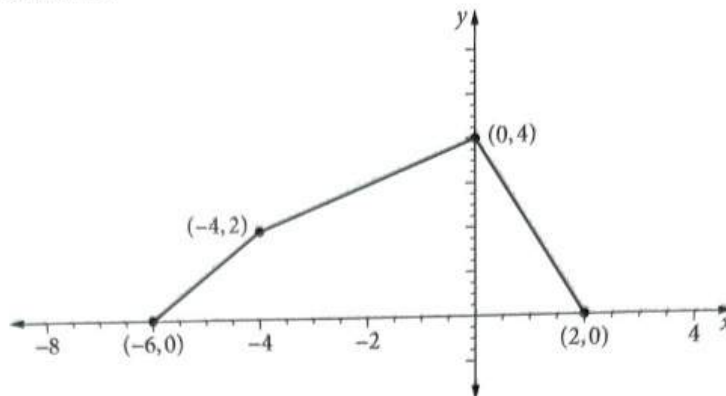


D

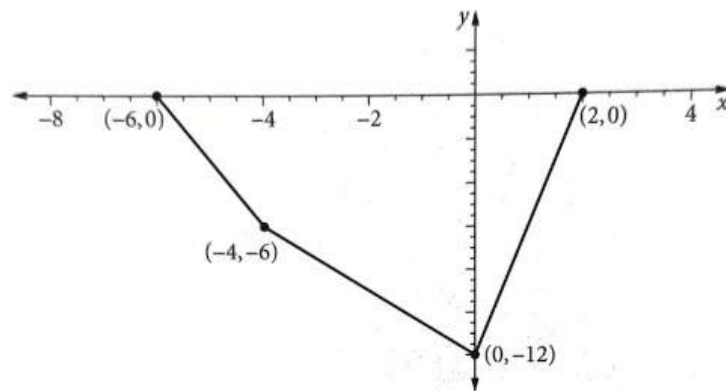


Question 10

The graph of $y = f(x)$ is shown.



The graph of $y = f(x)$ is transformed to $y = g(x)$ as shown.



Which of the following equations describes the relationship between $g(x)$ and $f(x)$?

- A $g(x) = 3f(-x)$
 B $g(x) = -3f(x)$
 C $g(x) = -f\left(\frac{x}{3}\right)$
 D $g(x) = f\left(-\frac{x}{3}\right)$

Question 11 ©NESA 2020 HSC EXAM, QUESTION 2

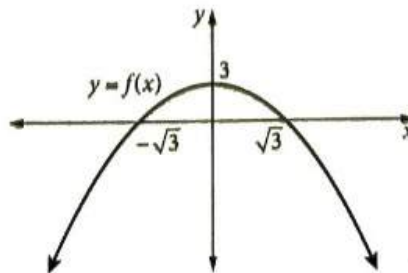
The function $f(x) = x^3$ is transformed to $g(x) = (x - 2)^3 + 5$ by a horizontal translation of 2 units followed by a vertical translation of 5 units.

Which row of the table shows the directions of the translations?

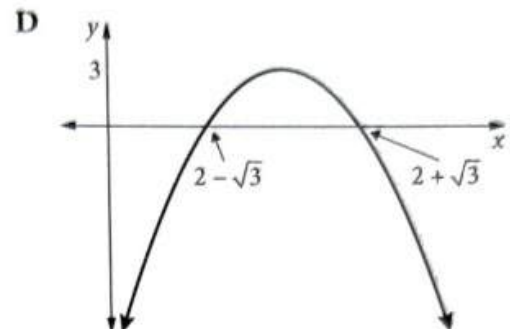
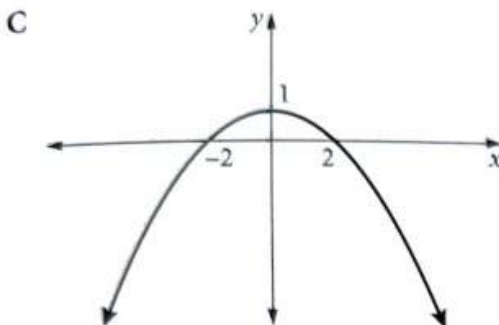
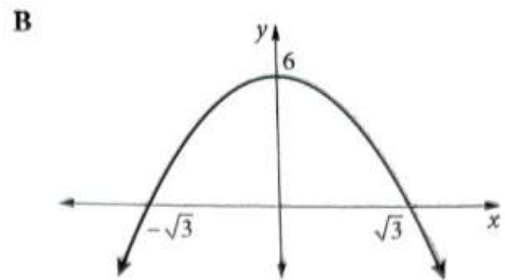
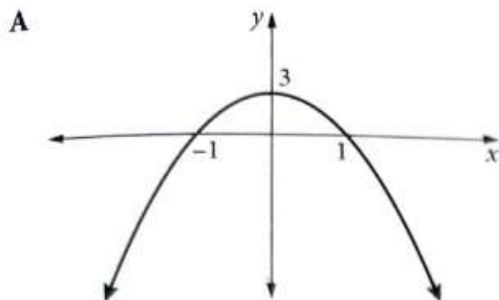
	Horizontal translation of 2 units	Vertical translation of 5 units
A	Left	Up
B	Right	Up
C	Left	Down
D	Right	Down

Question 12

The graph of $y = f(x)$ is shown.

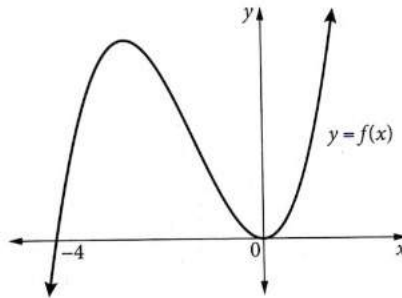


What is the graph of $y = 2f(x)$?



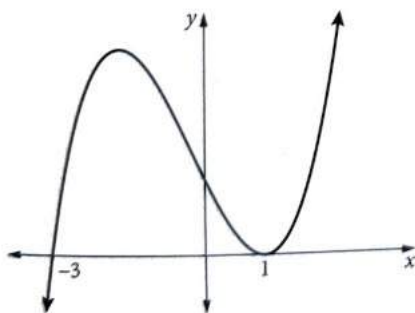
Question 13

The graph of $y = f(x)$ is shown.

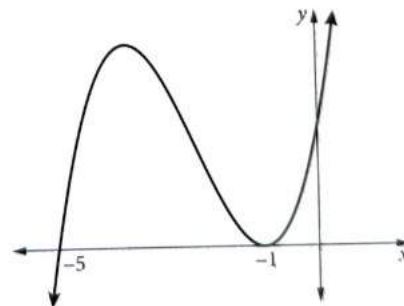


Which of the following shows the graph of $y = f(x + 1) + 1$?

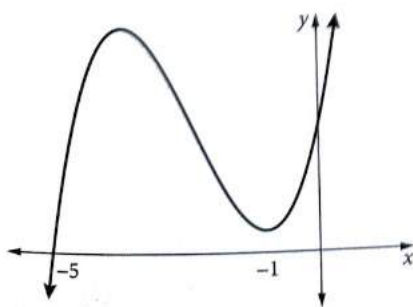
A



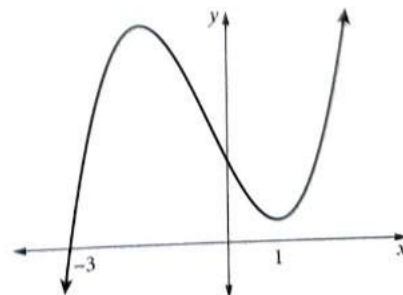
B



C



D



Question 14

The function $y = \log_e x$ is transformed by first being dilated vertically by a scale factor of 4 and then translated horizontally 3 units to the left. What is the equation of the transformed function?

- A $y = 4 \log_e x^3$
C $y = 4 \log_e (x - 3)$

B $y = 3 \log_e x^4$

D $y = 4 \log_e (x + 3)$

Question 15

$f(x) = (x - 1)^2$ is transformed to $g(x) = 2f(x + 4) - 5$.

What is the y -intercept of the transformed function?

A -5

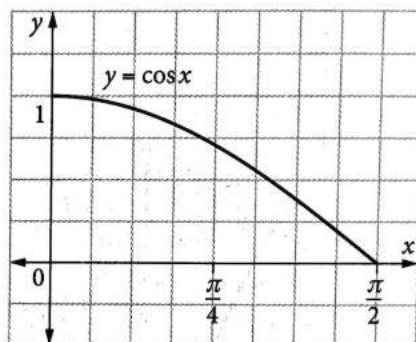
B -3

C 3

D 13

Question 16

The graph of $y = \cos x$ is shown below for the domain $\left[0, \frac{\pi}{2}\right]$.



Which equation, when graphed with $y = \cos x$ for this domain, would intersect it at exactly 2 points?

A $y = |x - 2|$

B $y = |x - 1|$

C $y = 1 - |x - 1|$

D $y = 1 - |x - 2|$

Question 17

The graph of $y = x^3 + x$ is reflected in the y -axis, then translated 1 unit to the right.

What is the equation of the new graph?

A $y = -(x + 1)^3 - (x + 1)$

B $y = (x + 1)^3 + x + 1$

C $y = -(x - 1)^3 - (x - 1)$

D $y = (x - 1)^3 - x + 1$

Question 18

Which equation represents the graph of $y = 2x^3 - 11x^2 - 5x + 8$ after it has been reflected in the y -axis?

A $y = 2x^3 + 11x^2 - 5x - 8$

B $y = 2x^3 + 11x^2 + 5x - 8$

C $y = -2x^3 - 11x^2 + 5x + 8$

D $y = -2x^3 - 11x^2 - 5x + 8$

Question 19

The graph of $y = f(x)$ undergoes three transformations in this order:

- a vertical dilation of a units
- a horizontal translation of b units to the left
- reflection in the x -axis.

What is the equation of the transformed graph?

A $y = af(x - b)$

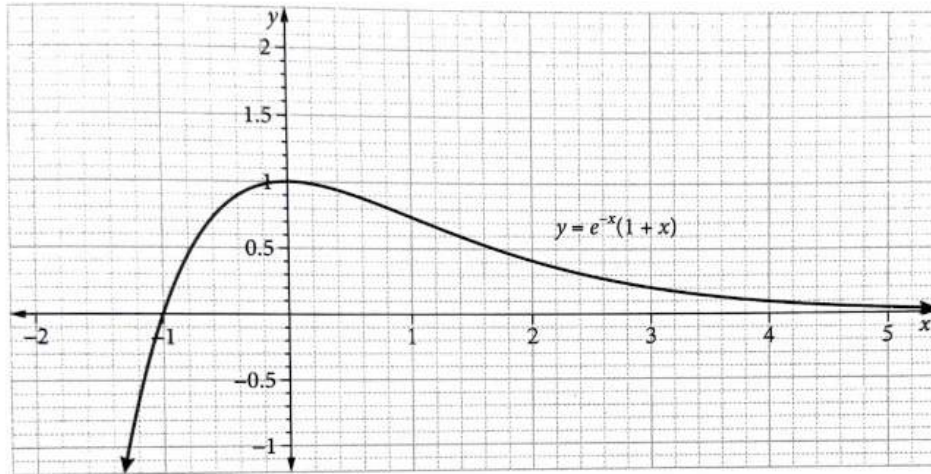
B $y = -af(x + b)$

C $y = -f(ax + b)$

D $y = f(ax - b)$

Question 20 ●●●

The diagram shows the graph of $y = e^{-x}(1+x)$.



How many solutions are there to the equation $e^{-x}(1+x) = 1-x^2$?

A 1

B 2

C 3

D 4

Graphing Techniques exam :

Q1) $y = f(x)$

$= f(x) - 2$

↳ graph moves down 2

$\therefore = (D)$

Q2) $(6, 2)$

4 units left reflected in x axis

$(2, -2) = (A)$

Q3) $y = \sqrt{x+2}$] left 2 units
 $= (D)$

Q4) $y = x^3 \rightarrow y = (x-5)^3 + 2$

HT 5 unit right] = (A)

HT up 2 units]

Q5) $y = \sqrt{2x-3}$

$2x-3 \geq 0$

$2x \geq 3$

$x \geq \frac{3}{2} = (D)$

Q6) $f\left(\frac{1}{2}\right) + 4 \Rightarrow (-3, 8)$

$= (-9, 12) = (C)$

Q7) $y = k(x+b)+c$

$b = -4$

$c = 6$

$k = -2$

(C)

Q8) $y = \frac{5}{x+3} \rightarrow$ 4 unit right, VD by $\frac{1}{2}$

$y = \frac{5}{x-1} \rightarrow (x+3) - 4$

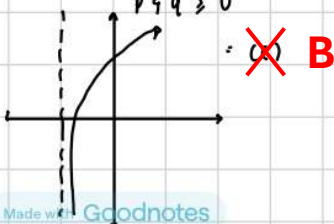
$= \frac{5}{x-1} \times \frac{1}{2}$

$= \frac{5}{2(x-1)} = (C)$

Q9) $f(x) = \log_3(x)$

$f(x) = \log_3(x-p) + q$

$p \neq q \geq 0$



Q10) $\begin{bmatrix} (-6, 0) \\ (-4, 2) \\ (0, 4) \\ (2, 0) \end{bmatrix}$

original

$\begin{bmatrix} (-6, 0) \\ (-4, -6) \\ (0, -12) \\ (2, 0) \end{bmatrix}$

transformed

VD SF 3

reflection in x-axis

$= g(x) = -3f(x) = (B)$

Q11) $f(x) = x^3$

$g(x) = (x-2)^3 + 5$

$= (B) \text{ (right 2, up 5)}$

Q12) original :

$(0, 3)$

$y = 2f(x) = (0, 6)$

$(-\sqrt{3}, 0)$

$(-\sqrt{3}, 0) = (B)$

$(\sqrt{3}, 0)$

$(\sqrt{3}, 0)$

Q13) $f(x+1) + 1$

left 1, up 1

$= (C)$

Q14) $y = \log_e x$

VD of SF 4

HT 3 left

$y = 4 \log_e(x+3) = (D)$

Q15) $f(x) = (x-1)^2$

y-int = $(0, 1)$

$g(x) = 2f(x+4) - 5$

$= (-4, 1)$

Help Q16, Q18

$(-4, 2)$

$(-4, -3) = (B)$

Q17) $y = x^3 + x$

$= (-x^3) + (-x)$

$= (-x^3-1) + (-x-1)$

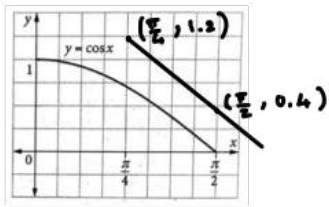
$= -(x^3+1) + -(x+1)$

$= (C)$

Q19) (B)

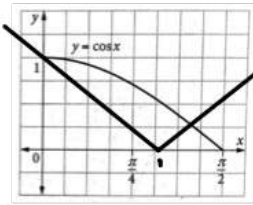
Q20) (B)

16. a). $y = |x-2|$



no solution

b). $y = |x-1|$



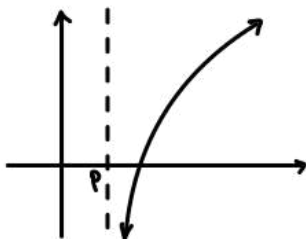
2 solutions

∴ Answer = (B)

9). $f(x) = \log_3 x$

$f(x) = \log_3(x-p) + q$, $p, q \geq 0$

∴ Shift p units right
and q units up



∴ (B)

17). $y = x^3 + x$

Reflect in y -axis: $y = (-x)^3 + (-x)$
 $y = -x^3 - x$

Translate 1 unit right: $y = -(x-1)^3 - (x-1)$

©

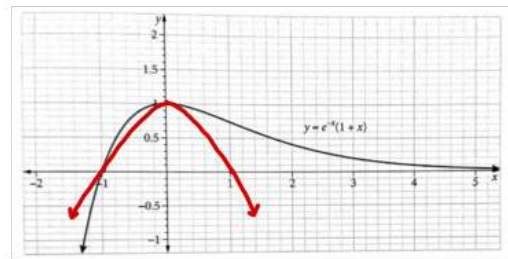
18. $y = 2x^3 - 11x^2 - 5x + 8$

Reflect in y -axis:

$y = 2(-x)^3 - 11(-x)^2 - 5(-x) + 8$
 $= -2x^3 - 11x^2 + 5x + 8$

∴ ©

20.



2 solutions

∴ (B)